

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain spiral model and iterative waterfall for software life cycle and discuss various activities in each phase.	Understand	6
B) What is the use of software development process models?	Understand	6
C) Describe about requirements validation and management.	Understand	6
Q.2 Solve Any Two of the following.		12
A) Write a difference between functional and non-functional requirements.	Analysis	6
B) What are the Extreme Programming Advantages?	Understand	6
C) What are the Characteristics of Agile software development?	Understand	6
Q. 3 Solve Any Two of the following.		12
A) Explain about the importance of sequence and activity diagrams in system design.	Understand	6
B) What is the use of an Activity diagram and draw an activity diagram for withdrawing money from a Bank ATM.	Apply	6
C) What is Use case diagram in UML? Draw use case diagram for an Online Shopping System.	Apply	6
Q.4 Solve Any Two of the following.		12
A) Explain about software architecture with suitable example.	Understand	6
B) What is Pattern Designing? Explain Characteristics of Pattern Designing.	Understand	6
C) Explain Structural design patterns. Make the list of Types of Structural Design Patterns and explain any three of them.	Understand	6
Q. 5 Solve Any Two of the following.		12

A) What are dependability properties in software engineering?	Understand	6
B) What is development testing? Write its advantages and disadvantages.	Understand	6
C) Write the differences between black box testing and white box testing.	Analysis	6

*** End ***